

MAJOR PROJECT REGISTRATION FORM

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	B07
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Design and Performance Analysis of an Optimized Reversible Quantum Adder cum Subtractor Using Qiskit

Abstract

Quantum computing is an emerging computing paradigm that utilizes the principles of quantum mechanics, such as superposition and entanglement, to perform computations more efficiently than classical computers for certain classes of problems. Unlike classical circuits, quantum circuits must be reversible to preserve quantum information and minimize energy dissipation. Reversible arithmetic circuits play a vital role in quantum computing systems, as they form the foundation for implementing arithmetic operations in quantum processors and quantum arithmetic units.

Addition and subtraction are fundamental arithmetic operations required in almost every computational application. Conventional quantum circuit designs often implement separate circuits for addition and subtraction, resulting in increased quantum cost, higher gate count, greater circuit depth, additional ancilla qubits, and more garbage outputs. These limitations reduce the efficiency of quantum circuits and make their implementation more challenging on current Noisy Intermediate-Scale Quantum (NISQ) devices, which have limited qubits, high error rates, and restricted connectivity.

This project proposes the design and implementation of an optimized **Reversible Quantum Adder cum Subtractor** using reversible quantum gates. The proposed circuit performs both addition and subtraction within a single configurable quantum circuit by utilizing a control qubit to select the desired arithmetic operation. The primary objective is to optimize the circuit by reducing quantum cost, gate count, circuit depth, ancilla qubits, and garbage outputs while ensuring the correctness and reversibility of all arithmetic operations.

The proposed design will be implemented using the **Python programming language** and the **IBM Qiskit** framework. The quantum circuit will be simulated using the **Qiskit Aer Simulator**, and its functionality will be verified through quantum simulation. The performance of the proposed reversible quantum adder cum subtractor will be evaluated using metrics such as quantum cost, gate count, circuit depth, qubit utilization, execution time, and circuit efficiency. The obtained results will be compared with existing reversible quantum adder-subtractor designs to analyze the improvements achieved by the proposed approach.

The expected outcome of this project is a resource-efficient and optimized reversible quantum adder cum subtractor that performs arithmetic operations accurately while utilizing fewer quantum resources.

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Base Paper:

[1] H. A. Bhat, F. A. Khanday, and B. K. Kaushik, "Optimized Quantum Implementation of Novel Controlled Adders/Subtractors," Quantum Information Processing, vol. 22, Article 174, 2023, doi: 10.1007/s11128-023-03896-4.

Supporting Papers:

[2] N. Mashayekhi, M. R. Reshadinezhad, and S. Moghimi, "A High-Speed Quantum Reversible Controlled Adder/Subtractor Circuit," 2024 15th International Conference on Information and Knowledge Technology (IKT), pp. 96–100, 2024, doi: 10.1109/IKT65497.2024.10892797.

[3] A. Kulkarni, M. Haghparast, and B. K. Kaushik, "Optimization and Performance Investigation of 1-Toffoli Gate Quantum Full Adders for Spin-Torque-Based n-Qubit Architecture," Optical and Quantum Electronics, vol. 56, Article 14, 2024, doi: 10.1007/s11082-023-05597-9.

[4] P. K. Kadbe and S. D. Markande, "Efficient Design of Reversible Adder and Multiplier Using Peres Gates," Applied Sciences, vol. 14, no. 20, Article 9385, 2024, doi: 10.3390/app14209385.

[5] J.-C. Jeon, "Multi-Layer QCA Reversible Full Adder-Subtractor Using Reversible Gates for Reliable Information Transfer and Minimal Power Dissipation on Universal Quantum Computer," Applied Sciences, vol. 14, no. 19, Article 8886, 2024, doi: 10.3390/app14198886.

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